Aftermath (War-torn City / Destruction-Focused Combined Arms)

Executive Summary

Aftermath is a **war-torn urban combined-arms battlefield** built for **heavy destruction, multi-domain coordination, rubble-based maneuvering, and autonomous adaptation to collapsing terrain**. The environment features shattered skyscrapers, cratered boulevards, burned-out industrial zones, fractured bridges, improvised fortifications, underground transit ruins, and smoke-filled avenues. The map emphasizes Obsidian Protocol's autonomous combined-arms logic, sensor-fusion under smoke and debris, comms-relay warfare, and logistics across unstable, partially collapsing terrain. In AR (Nreal Light/Air), the map anchors at ~4–5 m viewing distance, covering a 10x10 m physical area representing ~1 km² of war-torn urban terrain. Key features include destruction-driven chokepoints, multi-layer vertical combat, rubble-adaptive navigation, and combined-arms engagements across armor, infantry, drones, and artillery.

Design Goals & Gameplay

Destruction-Focused Combined Arms

Aftermath prioritizes armor pushes, drone reconnaissance, infantry clearing, artillery strikes, and air support. Terrain destruction directly reshapes tactical options.

Dynamic Urban Terrain

Buildings collapse, fires spread, dust clouds shift, and rubble piles grow. Autonomous units must constantly re-evaluate terrain viability.

Vertical & Interior Combat

Ruined towers, broken stairwells, collapsed floors, and underground transit tunnels create multi-layer combat zones.

Autonomy & Hazard-Aware Maneuvering

Units choose routes based on rubble stability, fire hazards, smoke density, and sensor clarity. Autonomous logic reroutes when terrain becomes impassable or collapses.

Sensor & Information Warfare

Smoke, dust, fire heat, metallic clutter, and collapsed structures degrade sensors. Relay towers on surviving rooftops restore clarity. Destroying relays creates blind pockets across the city.

Logistics & Resources

Fuel caches, industrial depots, and makeshift shelters serve as resupply nodes. Convoys must navigate cratered roads and avoid ambush-prone rubble corridors.

Deployment & Progress

A fixed deployment budget (~10,000 points) forces players to choose between heavy armor, urban infantry, drones, or artillery. Blue deploys from the southwest industrial ruins; Red deploys from the northeast fortified tower.

Spatial Constraints (Nreal AR)

Device Capabilities

Nreal Light/Air: ~53° diagonal FOV, 1920×1080 per-eye resolution. Optimal viewing distance ~4–5 m.

Playable Area & Scaling

10×10 m physical area ≈ 1 km² war-torn urban zone (1:100 scale). Designed for tabletop AR.

Tracking & Occlusion

Inside-out tracking detects horizontal planes. Depth-mesh (if available) provides limited occlusion. Virtual objects render over real-world view; assume minimal real-world occlusion.

Layout Overview

Coordinate System

1000×1000 grid (1 unit = 1 m). Origin (0,0) = southwest corner.

Key Locations

Blue HQ (Industrial Ruin Command) — (0,0), 150x150 m Burned-out factories, armored staging, drone pads
Red HQ (Fortified Tower Spire) — (1000,1000), 150x150 m Reinforced ruin with radar mast, AA emplacements, and sensor arrays
Central Ruin Basin — (500,450), 300x300 m Collapsed skyscrapers, shifting rubble, fires, dust clouds, multi-domain combat
Transit Tunnel Network — (250,800), 120x120 m Underground flanking routes, collapsed platforms, interior combat
Rubble Ridge Network — (800,300), 200x200 m High-ground overwatch created by collapsed structures; unstable but tactically valuable
Destruction Corridor — (0,450 → 400,900) Primary chokepoint; unstable rubble, fires, falling debris
Crater Boulevard Route — (300,0 → 450,350) Safer but slow; crater hazards and limited cover
Rooftop Relay A — (650,150), +240 m Long-range sensor vantage point
Rooftop Relay B — (150,650), +260 m Comms relay hub and hazard-monitoring node
Neutral Forward Outpost — (500,950) Drone resupply zone

Chokepoints & Sightlines

Destruction Corridor

Major chokepoint; terrain shifts unpredictably, ideal for ambushes.

Rubble Ridge Network

Interior flanking routes; unstable high-ground with long sightlines.

Ruin Basin

Combined-arms engagements broken by rubble; smoke reduces accuracy.

Crater Boulevard Route

Predictable but safe; vulnerable to overwatch from rubble ridges.

Sightlines

Rooftop Relay A provides long-range sensor coverage (~300 m). Smoke and dust create fog-of-war pockets.

Deployment Zones

Blue Deployment: SW industrial ruins + armor staging
Red Deployment: NE fortified tower + AA emplacements
Neutral Forward Zone: Outpost at (500,950) for drone drops

Assets & Environment

Terrain

Collapsed buildings, rubble fields, craters, broken roads, ruined depots.

Vegetation

Sparse weeds, dead trees, overgrown patches.

Buildings & Props

Military: AA guns, radar dishes, bunkers. Urban: Ruins, collapsed towers, vehicles, debris piles. Logistics: Trucks, crates, fuel drums, drones.

Effects

Smoke, dust, fire bursts, sparks, falling debris.

Audio

Distant explosions, collapsing rubble, machinery, radio chatter.

LOD & Budget

Asset Cat	Count	Tri/Item	Total Tris
Ruin/Rubl	3	200k	600k
Industrial	4	80k	320k
Buildings	15	50k	750k
Roads/Ra	4	30k	120k
Props	50	10k	500k
Total	—	—	**~2.29M**

Texture Budget

2048² for terrain/buildings, 1024² for props. DXT1/5 compression. Target ≤100 MB.

Unreal / Nreal Integration

Engine & Plugins

Unreal ARTemplate + Nreal SDK (XREAL). ARKit/ARCore enabled.

Rendering

Forward renderer, stationary lights, baked GI, aggressive culling.

Anchors

Single spatial anchor for entire map.

Interaction

Phone pointer for pan/zoom; HUD as screen-space widgets.

Multiplayer

Client-server architecture; offload AI/physics to server.

Implementation Plan & Testing

Milestones

1. **Prototype Layout — Month 1** Block-out ruin basin, rubble ridges, transit tunnels, AR anchor test
2. **Core Systems — Month 2** NavMesh, combined-arms autonomy behaviors, hazard logic
3. **AR Integration — Month 3** Anchors, plane detection, basic interactivity
4. **Content Fill — Month 4–5** Final models, textures, lighting, audio
5. **Gameplay & Polish — Month 6–7** Objectives, balance, VFX (smoke, dust, fire bursts)
6. **Testing & QA — Month 8** Multiplayer stress test, sensor validation

Testing Plan

Autonomy

Verify rubble navigation, combined-arms coordination, hazard avoidance.

Comms/Sensors

Test relay destruction, fog-of-war updates, LOS checks under smoke and dust.

Performance

≥60 Hz AR render on target device.

QA Checklist

Navigation, dynamic terrain logic, deployment budget enforcement, anchor drift, sensor accuracy.